

Math 164 Final Problem 7 Solutions

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Problem 7.

Question (a): Using cylindrical coordinates set up an integral for the volume of the region that lies within both the cylinder $x^2 + y^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 4$. DO NOT EVALUATE.

Solution: Recall the property that $Vol(E) = \iiint_E 1 dV$ where E is the region that wish to find the volume of. To use cylindrical coordinates, we must first represent z as a function in terms of $r \cos(\theta)$ and $r \sin(\theta)$. So, let $x = r \cos(\theta)$ and $y = r \sin(\theta)$. Then

$$z^2 = 4 - x^2 - y^2 = 4 - r^2$$

Hence, $z = \pm\sqrt{4 - r^2}$. So,

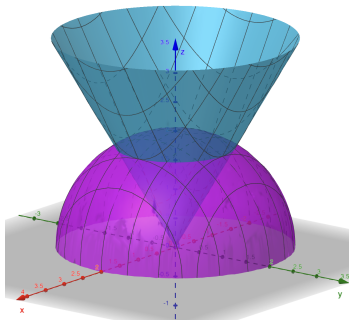
$$Vol(E) = \iiint_E 1 dV = \iiint_E r dz dr d\theta \tag{1}$$

$$= \int_0^{2\pi} \int_0^1 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r dz dr d\theta \tag{2}$$

Where the bounds for the first and second integral of equation (2) are obtained from simply observing that we should integrate from the origin to the edge of the cylinder.

Question (b): Using spherical coordinates set up an integral for the volume of the region bounded by the cone $z = \sqrt{3(x^2 + y^2)}$ and the hemisphere $z = \sqrt{4 - x^2 - y^2}$. DO NOT EVALUATE.

Solution: One should first be able to picture the cone and hemisphere, and the region bounded by them. They should have a similar picture as



Then, $Vol(E) = \iiint_E 1 dV = \iiint_E \rho^2 \sin(\phi) d\rho d\theta d\phi$. Observe that ρ ranges from $0 \leq \rho \leq 2$ and θ ranges from $0 \leq \theta \leq 2\pi$. Next we find the bounds for ϕ which will be given from the characteristics of the cone. So rewriting the equation of the cone we have

$$z^2 = 3x^2 + 3y^2.$$

Add $3z^2$ to both sides yields $4z^2 = 3(x^2 + y^2 + z^2)$. But recall from the distance formula that $\rho^2 = x^2 + y^2 + z^2$. So we can substitute ρ in and now we have $4z^2 = 3\rho^2$. Also, recall that $z = \rho \cos \phi$. So,

$$\begin{aligned} 4\rho^2 \cos^2 \phi &= 3\rho^2 \\ \cos^2 \phi &= \frac{3}{4} \\ \cos \phi &= \frac{\sqrt{3}}{2} \\ \phi &= \frac{\pi}{6} \end{aligned}$$

Thus, ϕ ranges from $0 \leq \phi \leq \frac{\pi}{6}$. Finally, we have our volume given by

$$Vol(E) = \int_0^{\frac{\pi}{6}} \int_0^{2\pi} \int_0^2 \rho^2 \sin(\phi) d\rho d\theta d\phi$$